

MURAT RAIMBEKOV

ML/CV Engineer

Bishkek, Kyrgyzstan | Open to remote

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Summary

ML Engineer specializing in tabular ML and computer vision. 39 merged PRs to Ultralytics YOLO, OpenCV, HuggingFace, PyTorch. Building production ML systems for real estate valuation. Strong foundation in PyTorch, gradient boosting, and feature engineering.

Experience

Junior Machine Learning Engineer

Jan 2026 – Present

Baker Tilly

Bishkek, Kyrgyzstan

- Building **Atlas** — automated real estate valuation platform for Kyrgyzstan market
- Fine-tuned **XGBoost**, **LightGBM** models achieving **R^2 0.76**, **MAE \$122/m²** on property valuation
- Built **NLP pipeline** using Gemini and Groq APIs for structured information extraction from listing descriptions

Data Science Intern

Jul 2025 – Dec 2025

Baker Tilly

Bishkek, Kyrgyzstan

- Performed data collection, cleaning, and feature engineering on **30,000+ property records**
- Built data pipelines for scraping and processing real estate listings from multiple sources

Open Source

39 merged PRs across Ultralytics YOLO (22), OpenCV (7), Transformers (6), torchvision (4)

Bug fixes (MoE CUDA non-determinism, OBB loss indexing, tv_tensors subclass preservation, empty masks handling), code refactoring, documentation

Listed in Ultralytics documentation authors

Projects

Real Estate Valuation (Central Asia) | *XGBoost*, *LightGBM*, *CatBoost*, *ResNet-50*, *Optuna* | [GitHub](#)

2025

- Built valuation models for Astana market achieving **R^2 0.83**
- Engineered **45+ features**: H3 geospatial tiles, POI distances, target encoding, spatial lag
- Built CV pipeline: ResNet-50 feature extraction → mean pooling → PCA on **64K property images**
- Published datasets and models on Kaggle and HuggingFace (**27K listings**, 64K images)

Technical Skills

Languages: Python, C++, SQL (PostgreSQL)

ML/CV: PyTorch, YOLO, OpenCV, Albumentations, timm, segmentation-models-pytorch, scikit-learn, XGBoost, LightGBM, CatBoost

Tools: Git, Optuna, pandas, NumPy, SHAP, Matplotlib

Education

American University of Central Asia

Bishkek, Kyrgyzstan

Bachelor's degree in Applied Mathematics and Informatics

Sep 2022 – May 2026

- Relevant Coursework: Neural Networks and Deep Learning, Neural Networks for NLP, Data Science and Machine Learning, Data Structures & Algorithms, Probability Theory and Statistics